Course 6: Present and Future of AI



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Present and future of AI

The present

- Assistants, reasoning models, **agents** acting on a computer,
- Models that generate **music**, images, video,
- Measured costs: **electricity**, money, dependence.

The future

- Not a prediction: **choices** being made now,
- By companies, states, artists… and engineers,
- Three open questions, each with **two sides**.

1. Work
Will AI destroy jobs?

2. Environment and sovereignty
Should France build its own frontier model?

3. Artistic expression
Generative AI: Instrument, or parasite for the artists?

The format

Each theme: a few slides of evidence, then **one** question with two sides. The next two sessions: you prepare, then you **debate**.

Outline

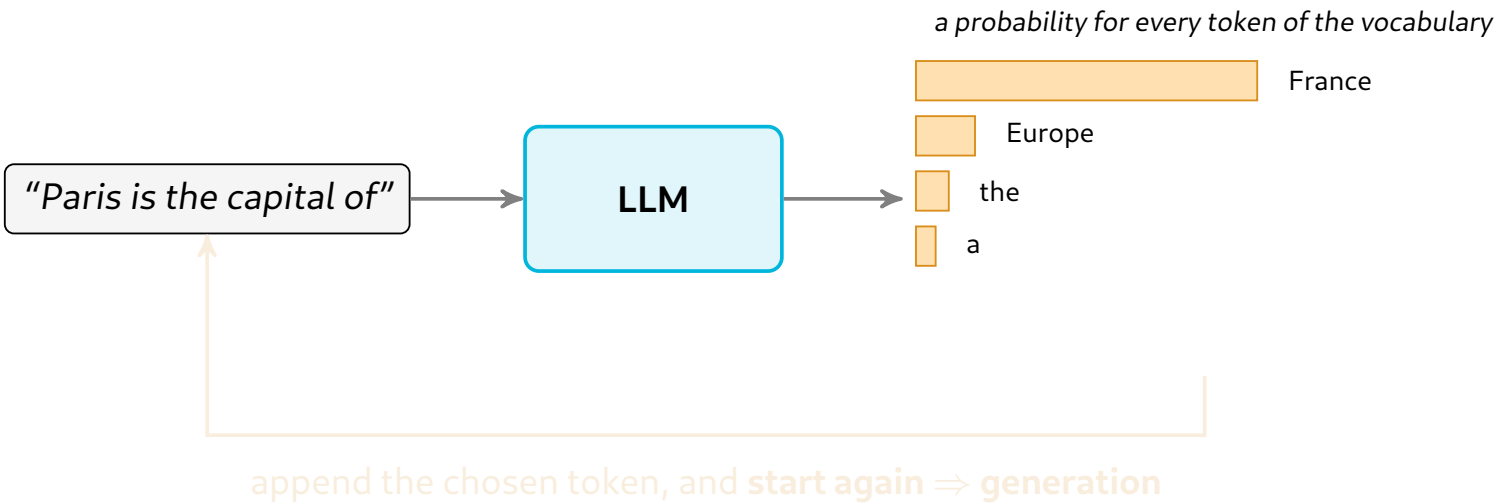
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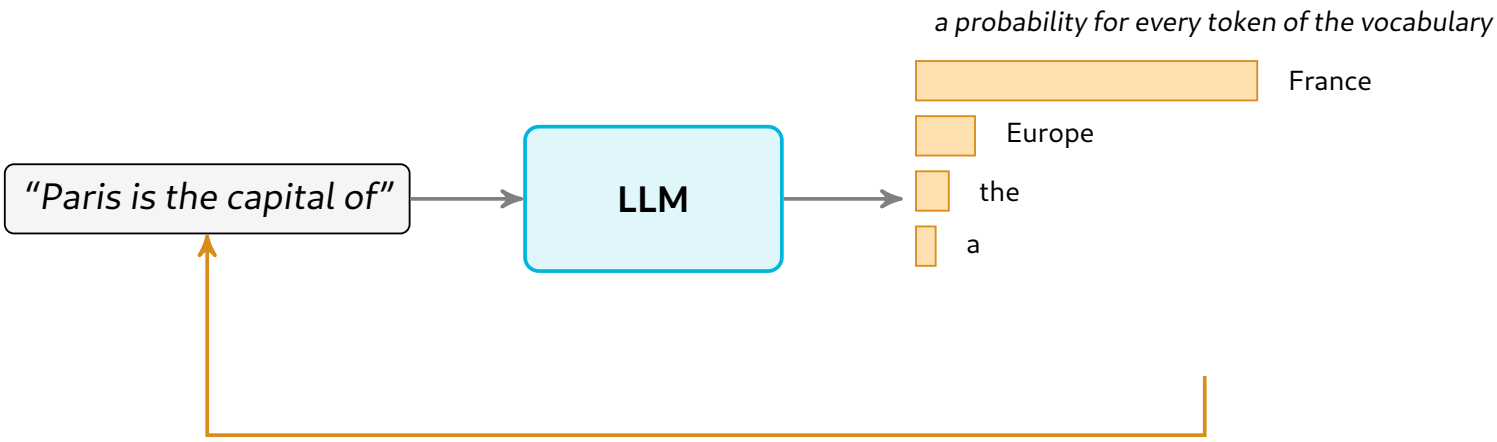
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4 Next sessions: the debate

Using a Large Language Model (LLM) to generate text



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From text completion to instruction following

The problem

Pre-training teaches the model to **continue text**, not to **answer**.

Before (raw pre-trained)

You: What is the capital of France?

Model: What is the capital of Germany? What is the capital of Spain? ...

it just keeps the pattern going

After (instruction-tuned)

You: What is the capital of France?

Model: Paris.

it answers the actual question

Base model

continues text

instruction tuning

supervised learning

on (instruction → answer) pairs

Assistant

follows instructions

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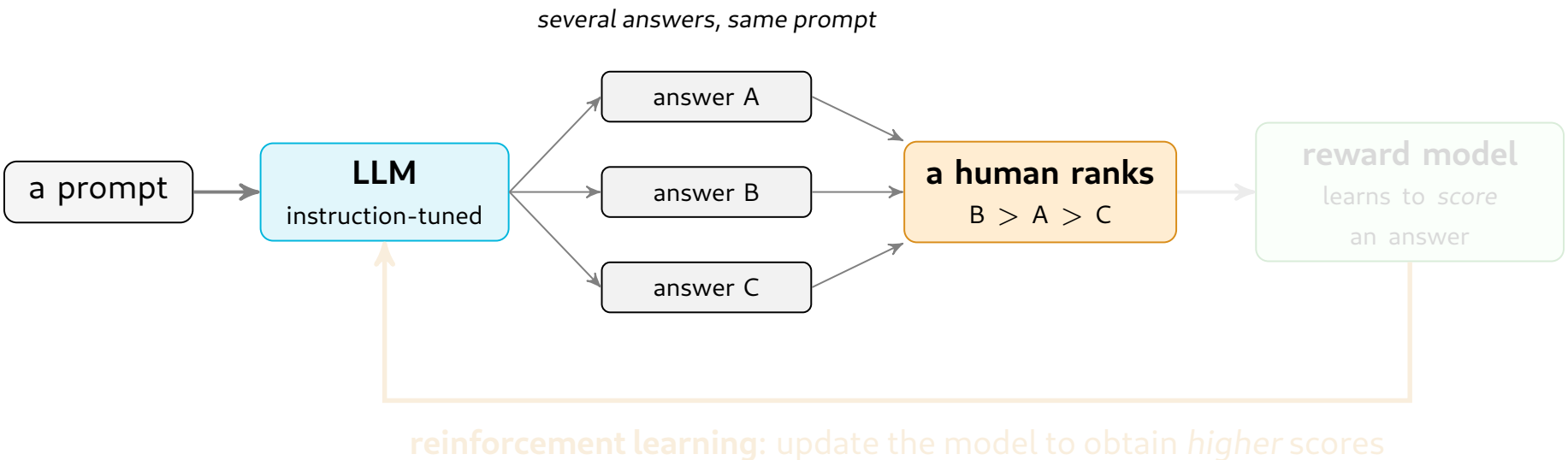
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Reinforcement Learning from Human Feedback

RLHF: Put human in the loop

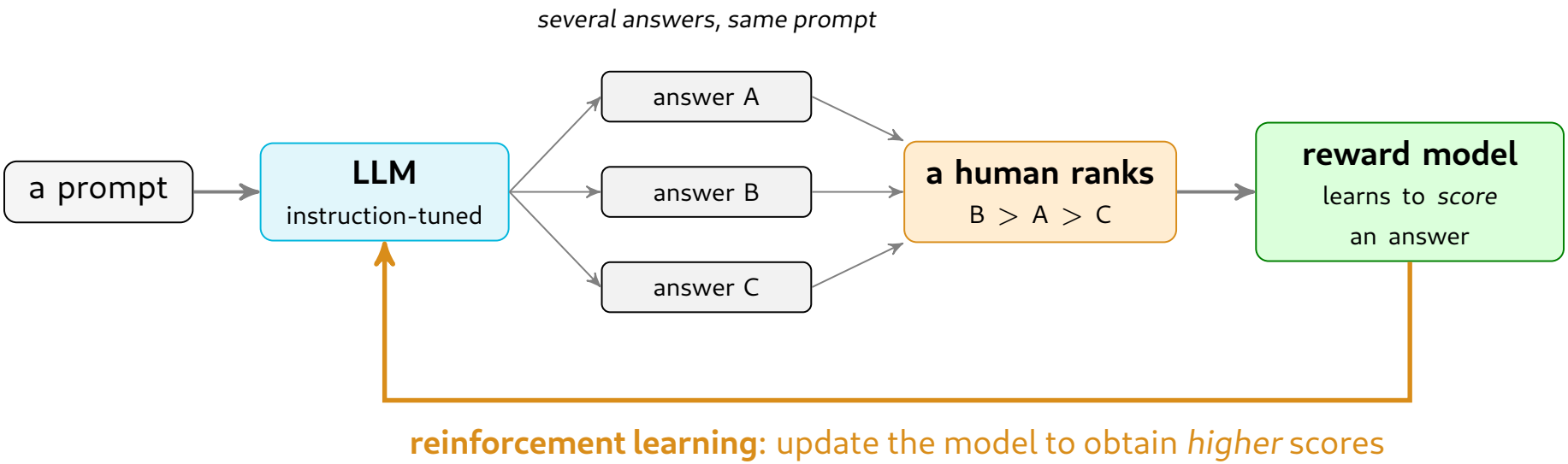
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Reinforcement Learning from Human Feedback

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Assistants are not just LLMs



RLHF and sycophancy

The model optimizes **what humans approve of**, based on pre-defined criteria.
Hence the behavior of some assistants: sympathic, enthusiastic, ... but sometimes wrong to please you ("sycophancy").

RLHF is subject to the will of people training it: they may also be used to make a model antipathic, aggressive, manipulative, or else!

Reasoning models: Chain-of-Thought

Giving time to think

Taking time to write a reasoning process before answering.
Leads to better answers on arithmetic, logic, and planning problems, but increases the computation cost.

Direct answer

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 43 ×

it guesses in one shot

Chain-of-thought

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 6 cans × 7 = 42;
5 + 42 = **47**. ✓

the steps first, then the answer

Chain-of-Thought largely participates in the recent progesses of LLMs!

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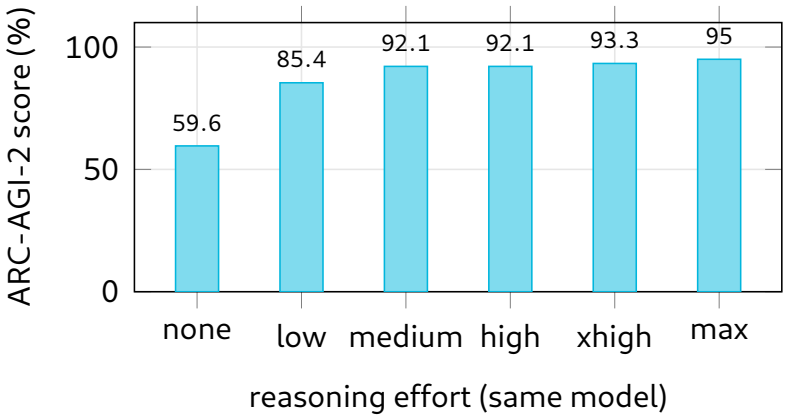
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CoT performance, and its cost

Two places to spend compute

Training: paid once, the model improves for every user.
Inference (CoT): paid at **every question**, and you choose how much.



Longer reasoning improves accuracy!

GPT-6 Astra: **+35 points**, no retraining.

It is also costly

- Paid at **every query**, plus **latency** and **energy**,
- Prices reduce with time (87.5% on ARC-AGI-1 cost **$4,560**/task (2024), **$0.30** (2026))...
- ... but increasing the performance requires larger CoT: +26 points (none → low), then only +10 (low → max) for 100× more compute.
- The frontier stays expensive: one ARC-AGI-3 run with GPT6-Astra costs **$26,098**.

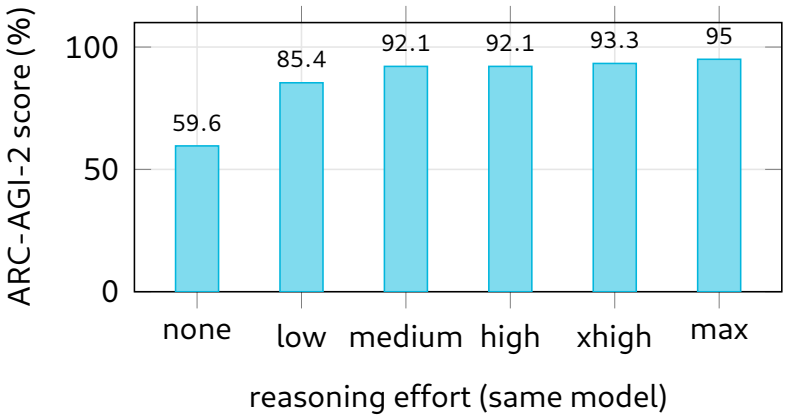
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ARC Prize Verified: GPT-6 Astra, Sep 2026, `https://arcprize.org/results/openai-gpt-6-astra` Grok 4.6 ($0.30), Aug 2026, `https://arcprize.org/results/xai-grok-4-6` o3 ($4,560), Dec 2024, `https://arcprize.org/blog/oai-o3-pub-breakthrough`

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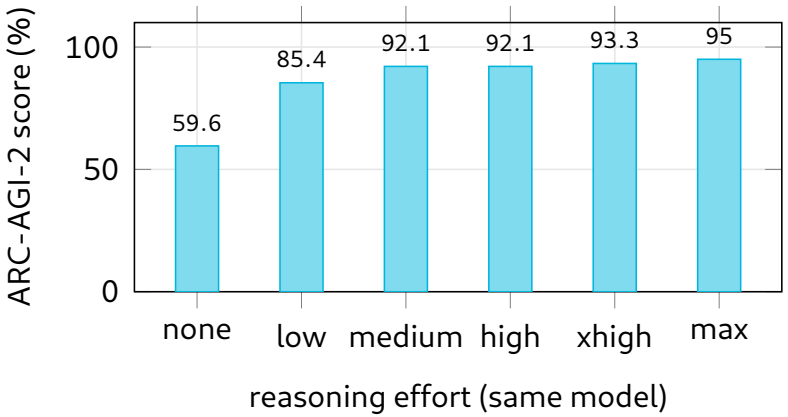
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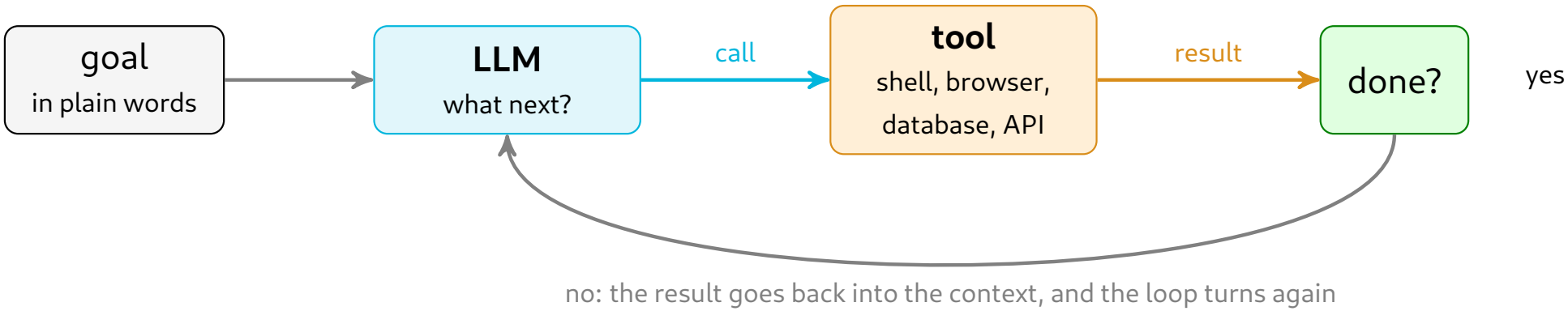
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Agentic AI: from answers to actions

Agents

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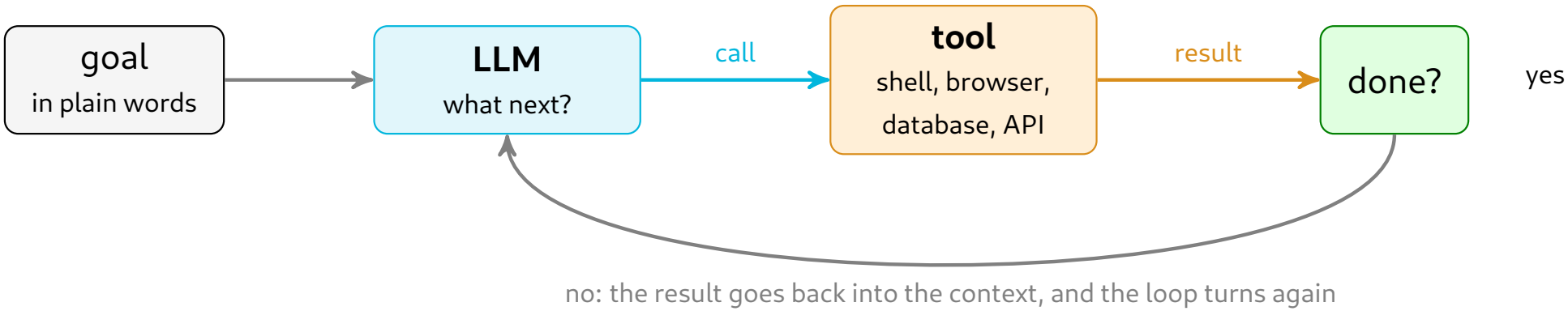


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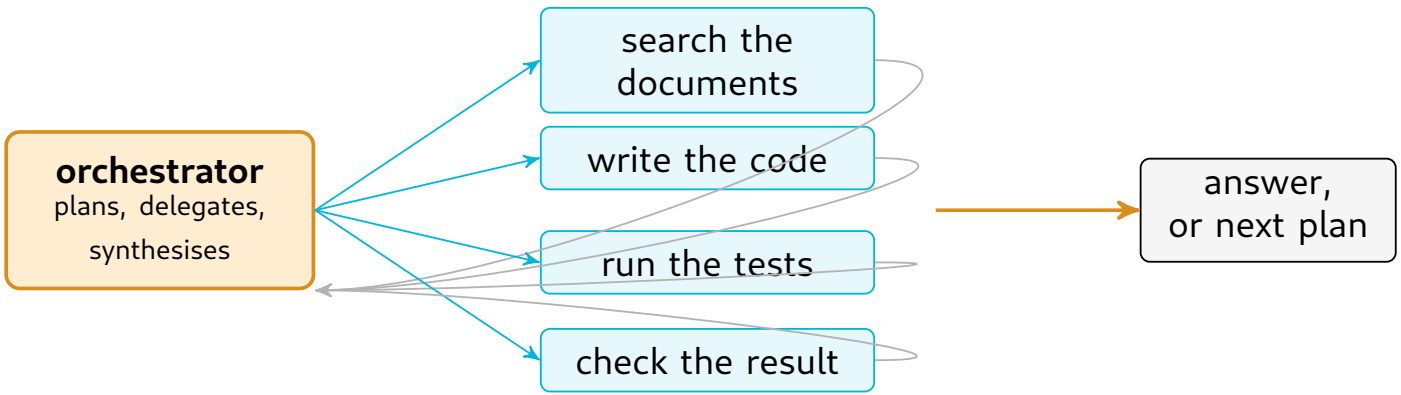
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The LLM is an orchestrator of many agents

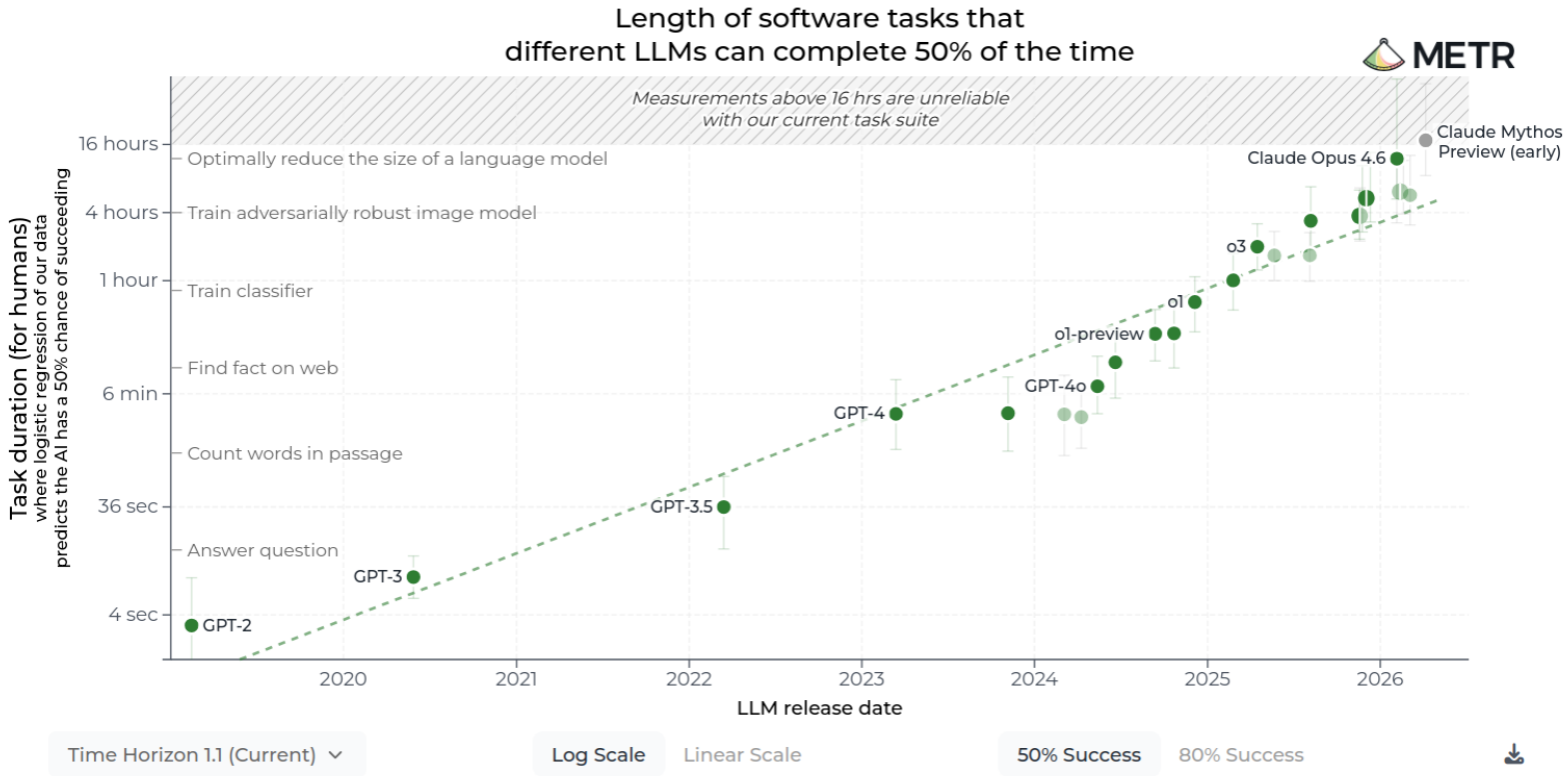
The LLM orchestrates subtasks, but does not take action itself.

1. The LLM understands the overall task and breaks it down into smaller, manageable subtasks.
2. It **delegates** the pieces (to other model instances, or to deterministic tools).
3. It **recombines** the results, and decides whether to continue or to return the final answer.

Notice that the subtasks are decided at run time, not written in advance.



Agents can perform longer and longer tasks



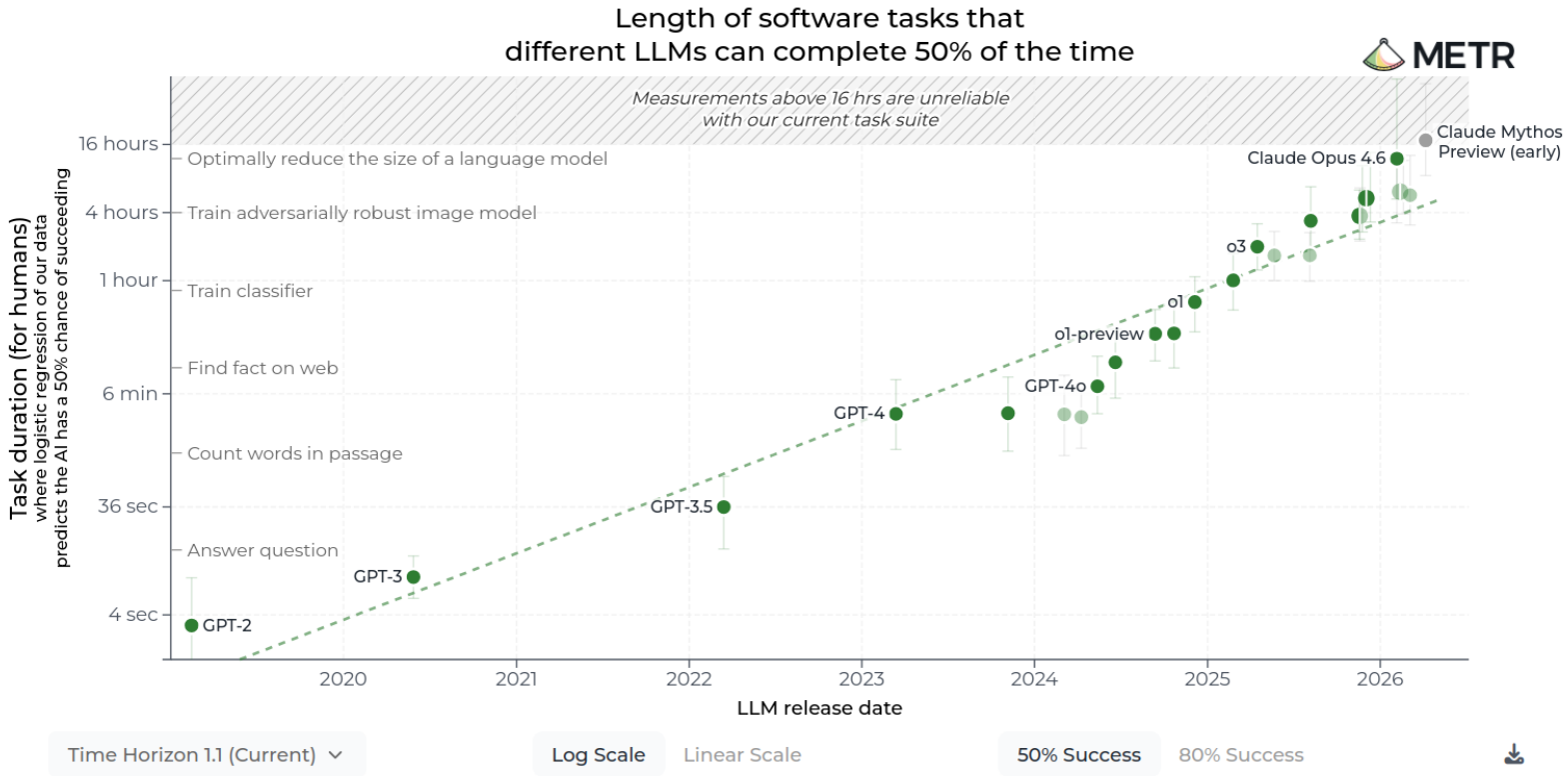
Read it as time saved using agents

2023: a few minutes of an engineer's work. / 2026: a **working day**, and more.
But 50% is not reliable: **someone still has to check**.

Models are progressing very fast $\Longrightarrow$ more and more useful.

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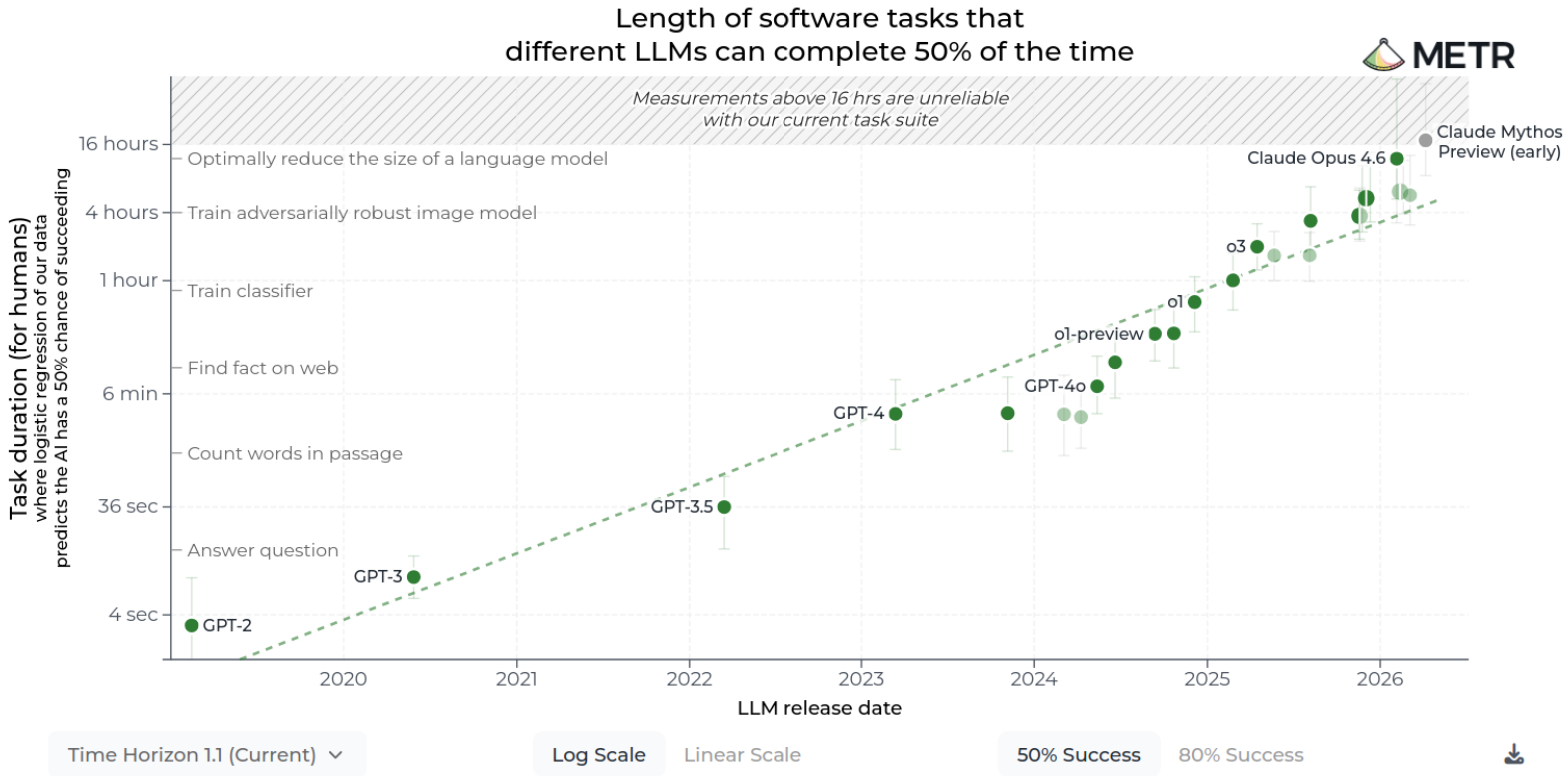
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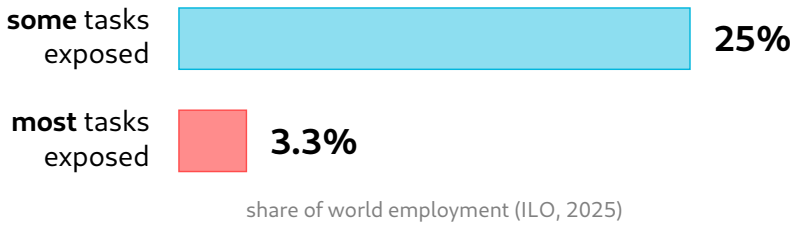
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What is the expected impact of AI on jobs?

1. Exposure

Which tasks of a job can AI do?



2. Productivity

How much faster with AI? (randomised trials)

+26% tasks done
developers, three firms

−40% time spent
professional writing

gains **largest for juniors**

Expected impact: AI transforms jobs more than it replaces them.

But: young people are hired less

US, most exposed jobs, employment vs. expected

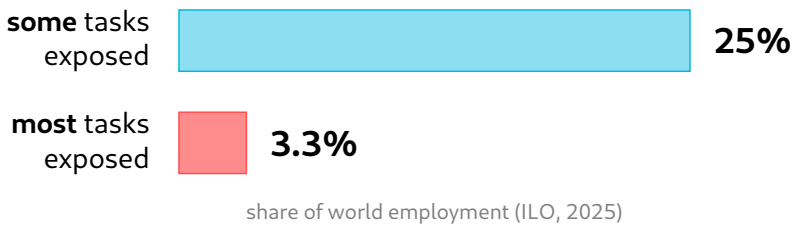
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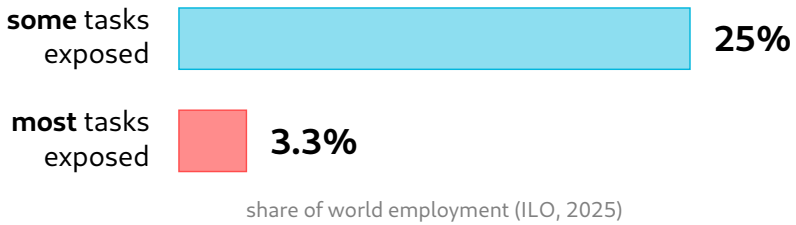
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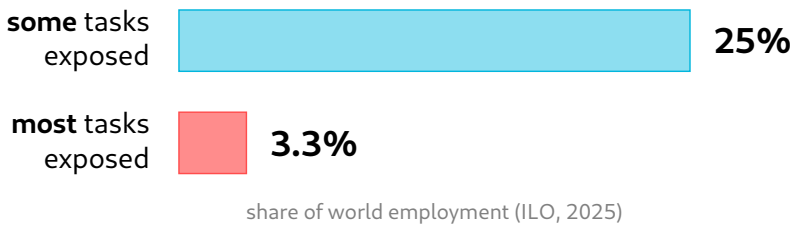
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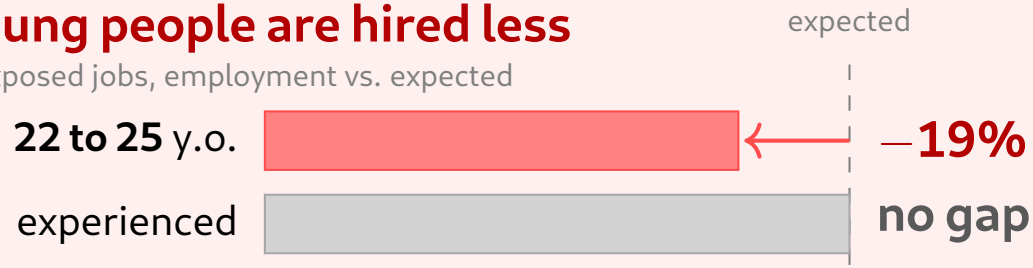
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Place yourself: will AI destroy jobs?

pick one side

YES *or* **NO**

Strongest for "yes"

- Task length doubles every $\approx$ **4 months** (12 h in 2026),
- Each answer has a **price per task**, falling fast (\$4,560 $\rightarrow$ \$0.30): compare it with a wage,
- Young people already hired less (-19%),
- No juniors today $\Longrightarrow$ no seniors tomorrow.

Strongest for "no"

- "Exposed" $\neq$ replaced: **3.3%** of jobs have most tasks exposed,
- Measured gains: **same person, faster** (+26% tasks),
- 50% success only: **someone still has to check**,
- 60% of 2018 jobs did not exist in 1940.

Autor et al., *QJE*, 2024 (jobs of 1940 and 2018) other sources: previous slides

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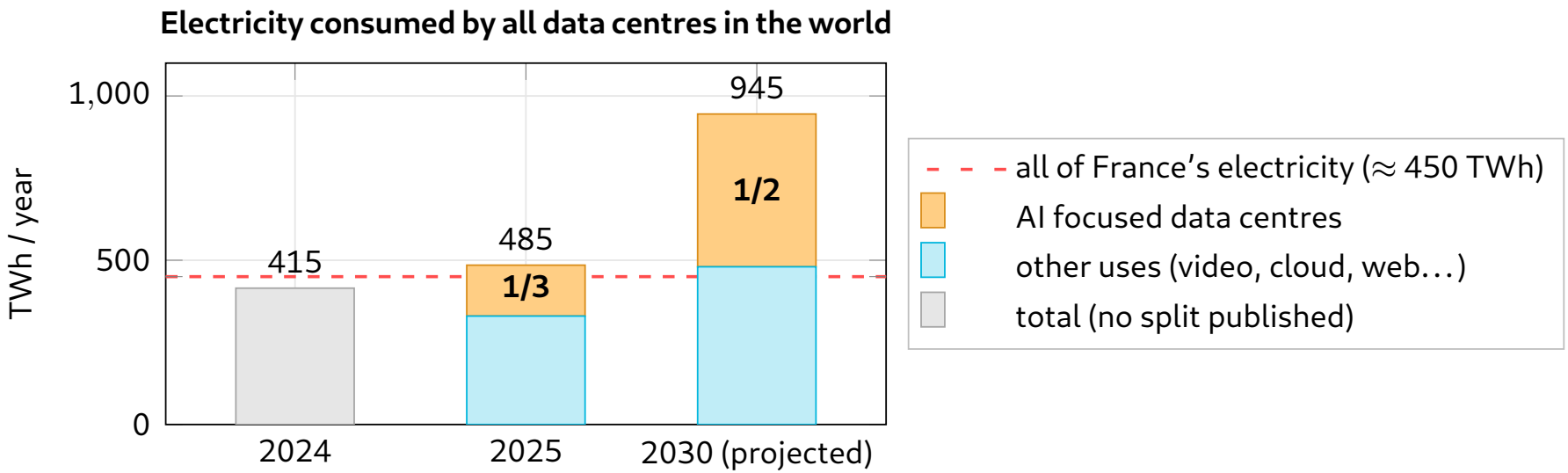
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(AI alone, 2025: ≈ 0.5%)

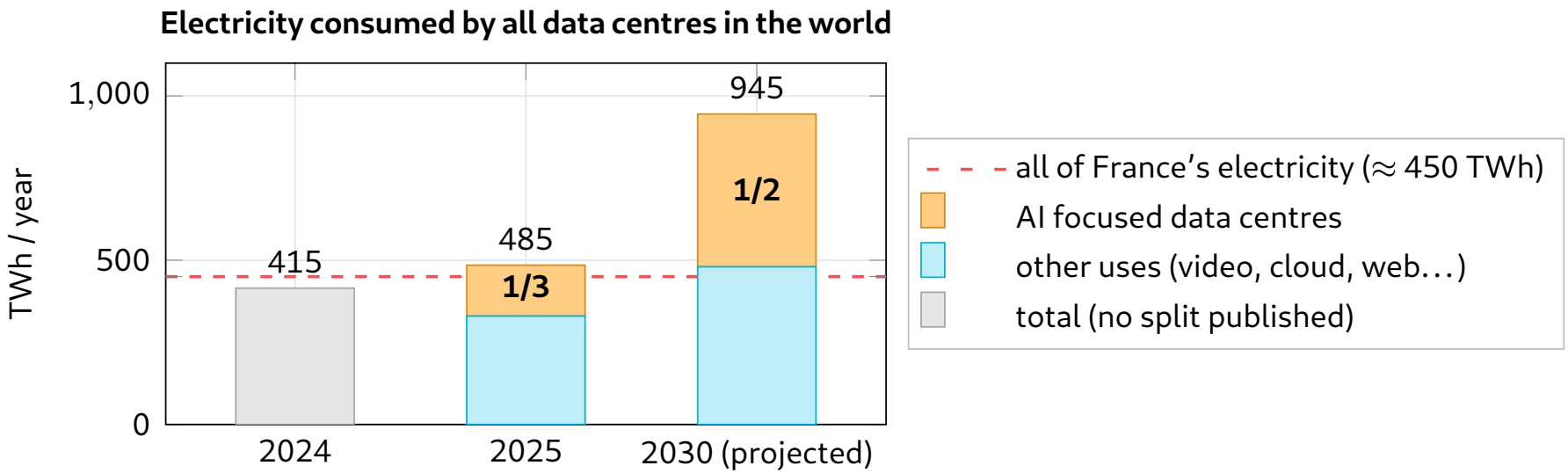
+17%
in 2025 (AI alone: +50%)
all electricity: +3%

×2
by 2030
(AI alone: ×3)

A small share today; but it's one of the fastest growing uses of electricity, and AI drives the growth.

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One model, measured: Mistral Large 2

Building the model
training, then 18 months of service

20,400 t CO_2e
≈ *the yearly footprint of 2,200 French people*

a large cost, for the model as a whole

+

One more answer
Le Chat, 400 tokens

1.14 g CO_2e

a small cost, at each use

Cost breakdown

Computing (training and inference) account for **85.5%** of the emissions. Mistral does not publish the split between training and inference.
The rest: making the hardware and the buildings.

Training and using one Mistral model costs as much as a small town…
And we don't know for OpenAI, Anthropic, Google, Meta, etc.

Mistral AI, Carbone 4 and ADEME, 22 Jul 2025,
`https://mistral.ai/fr/news/our-contribution-to-a-global-environmental-standard-for-ai/` Equivalences computed with ≈ 9 t CO_2e per French person and per year (SDES) and 2,500 m^3 per Olympic pool

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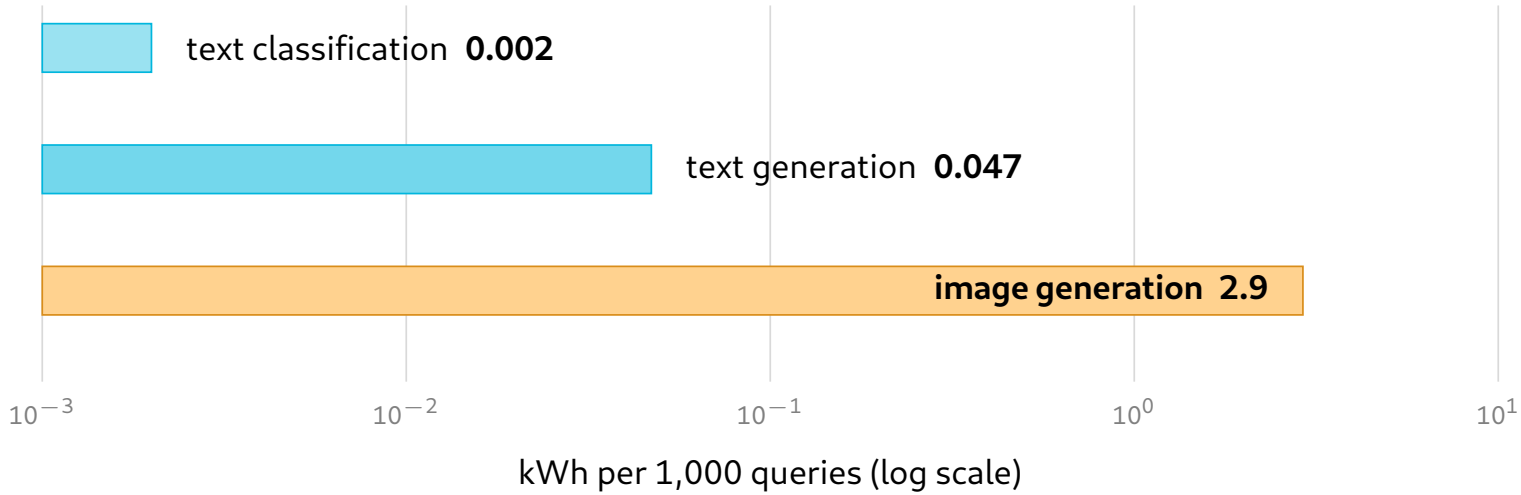
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Not all tasks cost the same



× 1,450 between the cheapest and the most expensive task

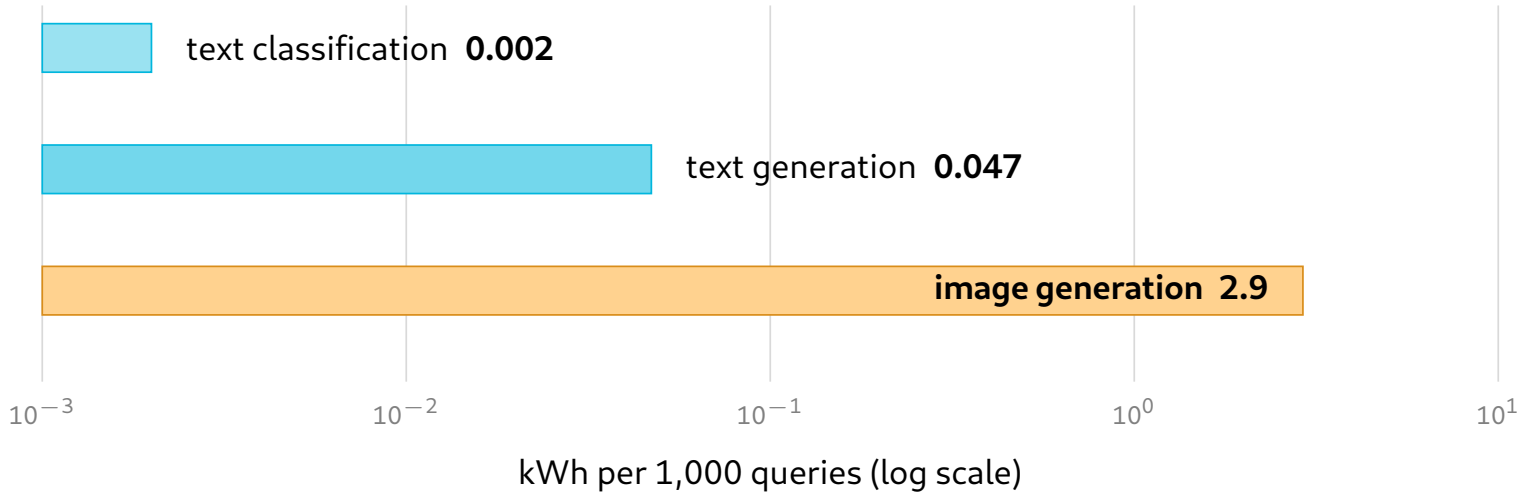
And reasoning models (first part of this course)

A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

"Using AI" is not one activity.

Luccioni, Jernite, Strubell, *Power Hungry Processing*, FAccT 2024, `https://arxiv.org/abs/2311.16863` Oviedo et al., *Joule*, Apr 2026 Google, `https://arxiv.org/abs/2508.15734`

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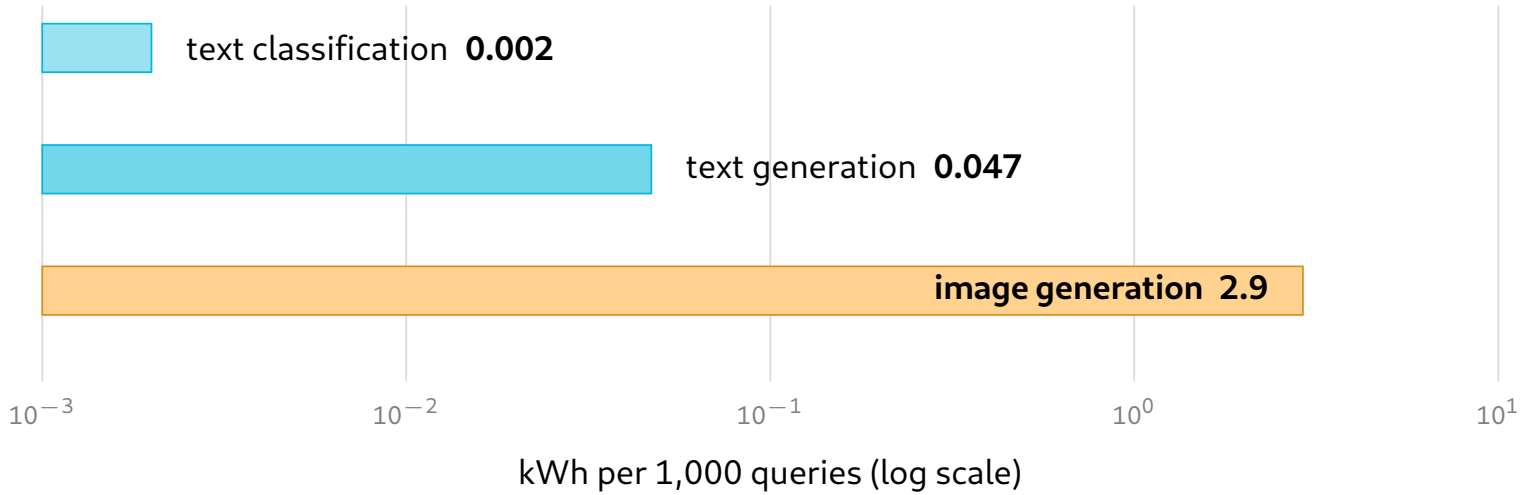
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A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

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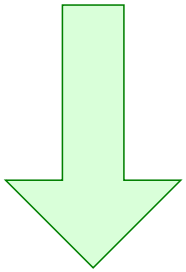
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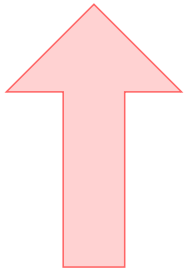
Computing efficiency, and the rebound effect



÷ 33

energy **per prompt** at Google, in 12 months

but



+37%

Google **total** electricity, in 2025

(emissions: +81% since 2019)

Rebound effect (Jevons paradox)

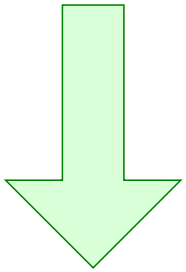
"It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth."
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So: developing and running models is costly

Running: cheaper answers → more answers, in more products, for more people.
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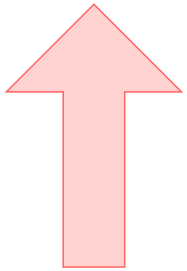
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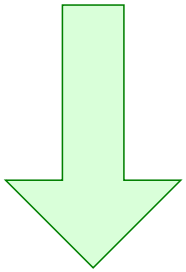
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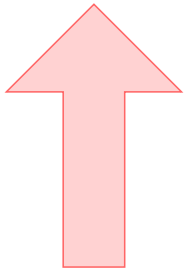
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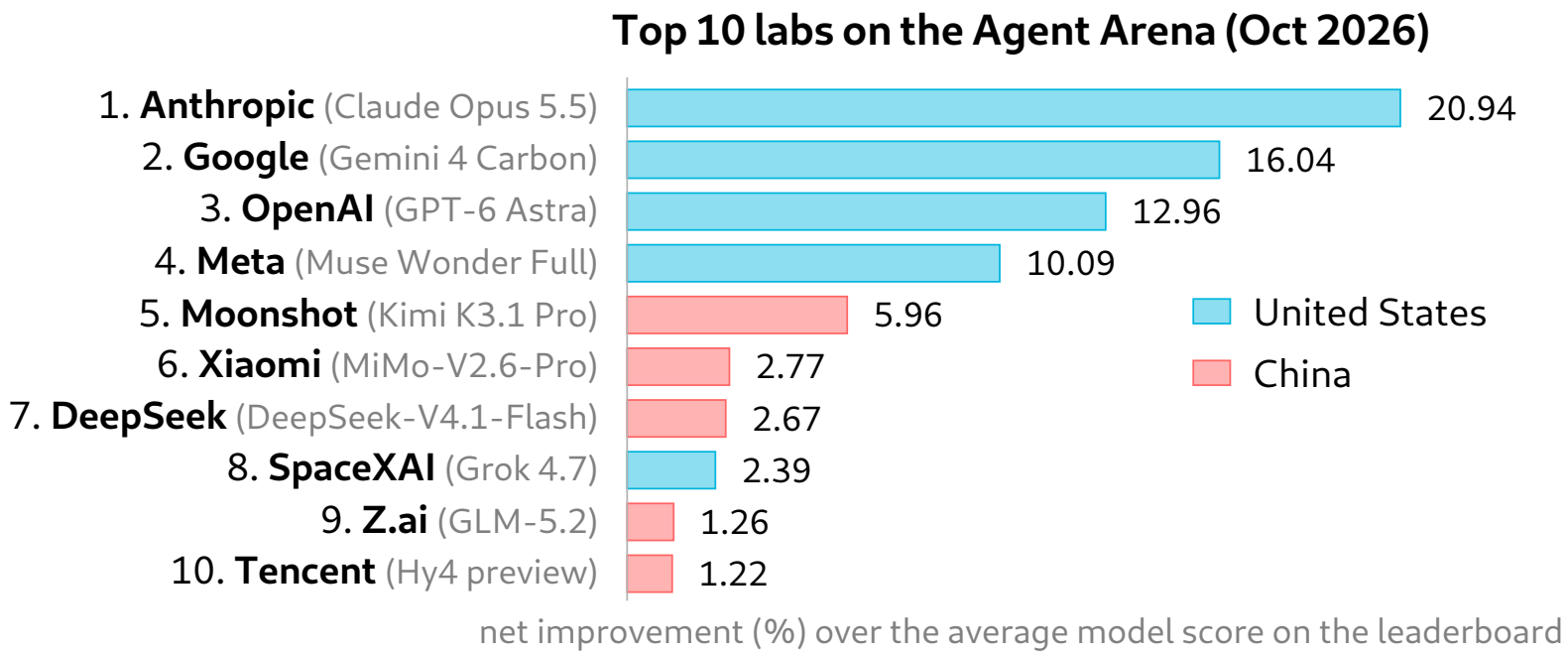
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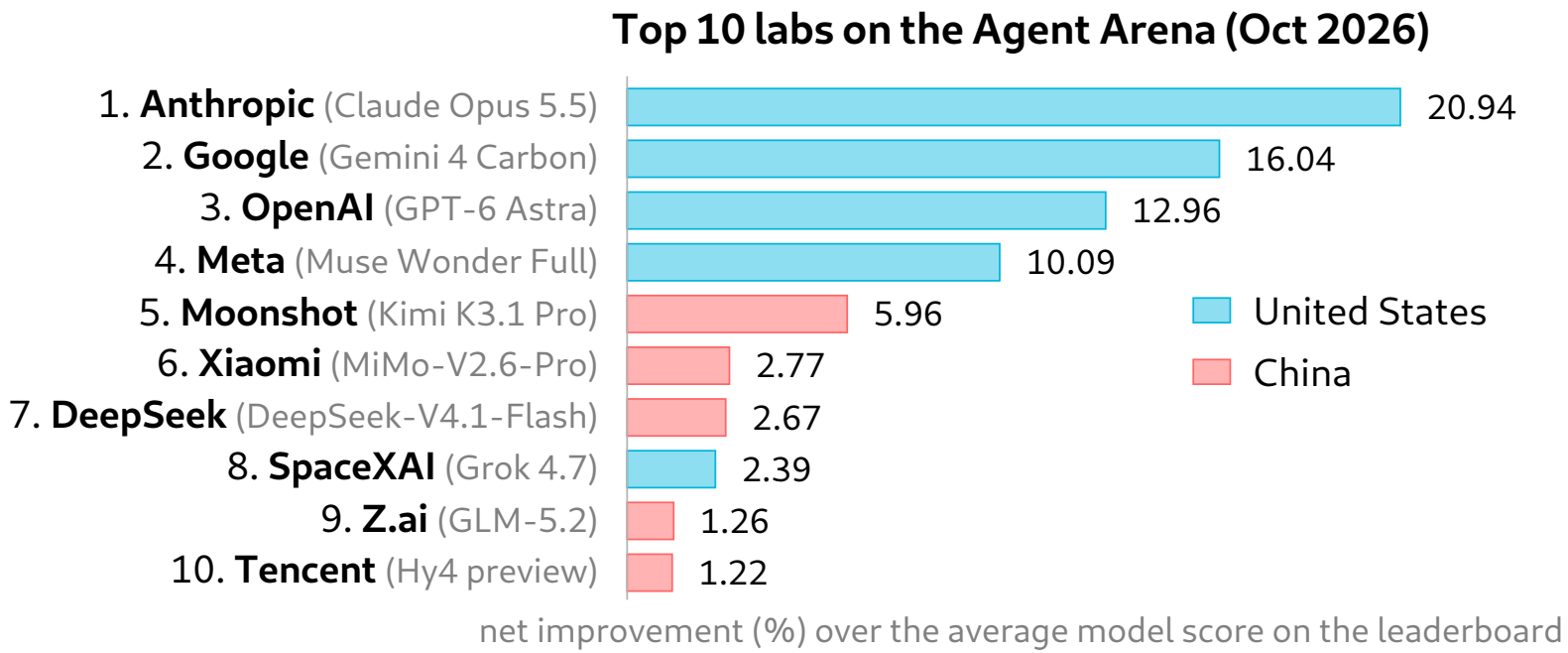
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Top 10: 5 American labs, 5 Chinese labs.
No European lab.

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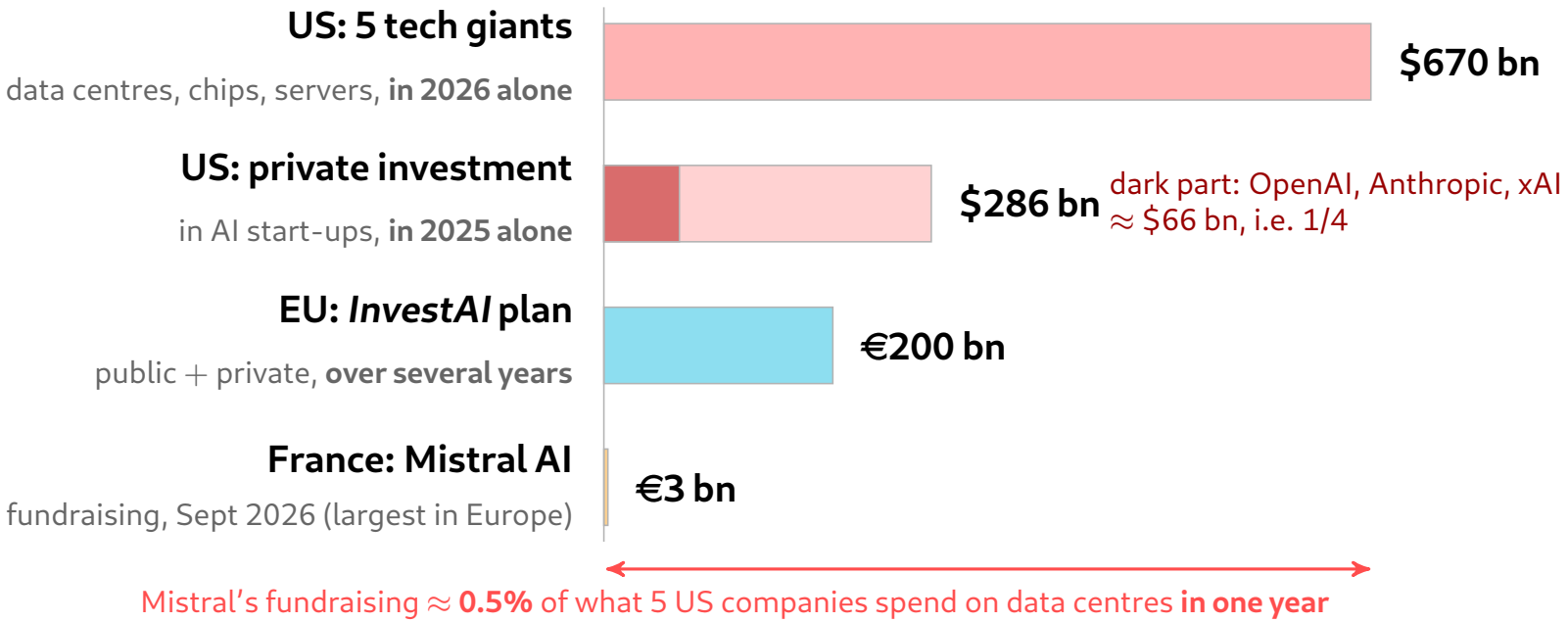
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A ferocious and very costly competition



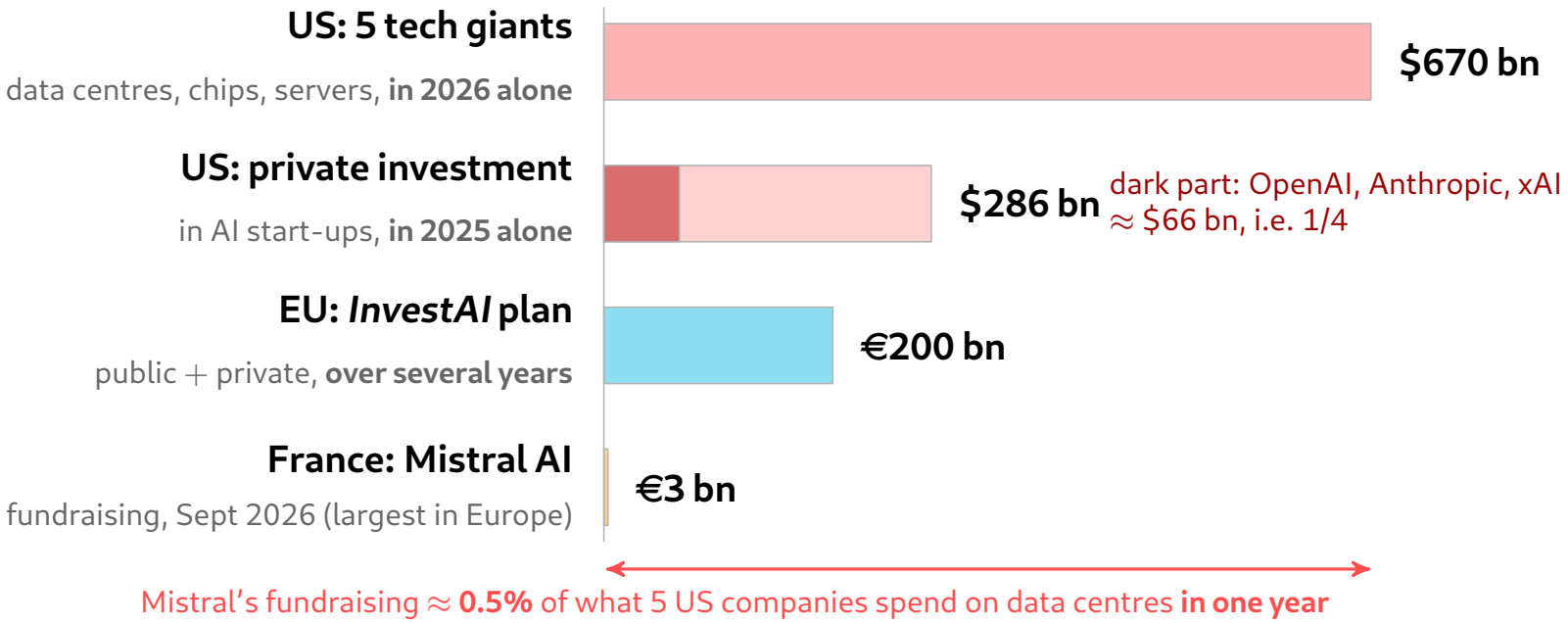
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France (and even Europe) is left behind (for now).

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Sovereignty is not only the AI model!

link	who dominates the world	and Europe?
fabrication	TSMC (Taiwan): **70%** of chip making[1]	**ASML** (NL): sole maker of advanced chip printing machines[2]
chips	NVIDIA (US): >**60%** of AI computing[3]	no European AI chip at scale
cloud	Amazon, Microsoft, Google: **63%**[4]	European providers: **15%** of Europe[5]
data centres	US **45%**, China 25% of world consumption[6]	Europe **15%**[6]; low carbon power in France
models	Agent Arena top 10: US **5**, China **5**[7]	Europe: **0** (Mistral not in the top 10)[7]

Five links **in series**

Independence requires all five. And four of them are **physical**: this industry looks like software, and behaves like heavy industry.

Perfect sovereignty would be very expensive.

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Already dependent: Europe runs on American Big Tech



Dependence also means loss of control

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And a foreign law, and a foreign president, can reach that switch.

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Place yourself: should France build its own frontier model?

pick one side

YES *or* **NO**

Strongest for "yes"

- Already dependent: **70%** of European cloud is American (CLOUD Act),
- The Internet lesson: European cloud **29%** $\rightarrow$ **15%**,
- Arena top 10: US **5**, China **5**, Europe **0**,
- Real assets: ASML, Mistral, low carbon power.

Strongest for "no"

- One model $\approx$ yearly CO_2 of **2,200** French people,
- Rebound effect: per prompt $\div$**33**, total **+37%**,
- Mistral raises **0.5%** of one year of US spending,
- A model is 1 link out of 5: still dependent.

Sources: previous slides

Outline

1 Agentic AI: Will AI destroy jobs?

2 Environment and sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate

Generating art from one sentence is now easy.

Who needed it? And who pays for it?



	AI as a tool: the human chooses		AI as a generator: the model chooses
images	restoration, upscaling, removing a background	→	a picture from one sentence (Midjourney)
text	spelling, translation, rephrasing	→	a whole story from a prompt
voice	noise removal, restoring old recordings	→	a cloned voice reading any text
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What AI music enables

the human makes the choices — **the model makes the choices**

Restore
The Beatles, *Now and Then* (2023)
Lennon's voice **separated** from an old cassette (separation, not generation). Grammy 2025

Tools for musicians
Stems, mastering, timbres
separate the instruments of a track, master a demo, turn your voice into a violin (RAVE, IRCAM, 2021)

Give a voice back
Randy Travis (2024)
voiceless after a stroke, releases a new song: his voice recreated by AI, **with his consent**

New creators
Xania Monet (2025)
a human writes the lyrics, Suno makes the music: $3M record deal. Suno: **2 million** paying users (2026)

Some music was never meant to be art

Jingles, video backgrounds, game loops: **cheap music** for small creators who could never pay a composer.

Some music was never meant to be serious

A song for a friend's birthday, a joke with your band: like karaoke, the point is **the fun of making**, not the art.

Different models and uses mean different ethical considerations.

Randy Travis, May 2024 Xania Monet, Sep 2025 Suno, Feb 2026 (company figure) RAVE: Caillon & Esling, `https://arxiv.org/abs/2111.05011`

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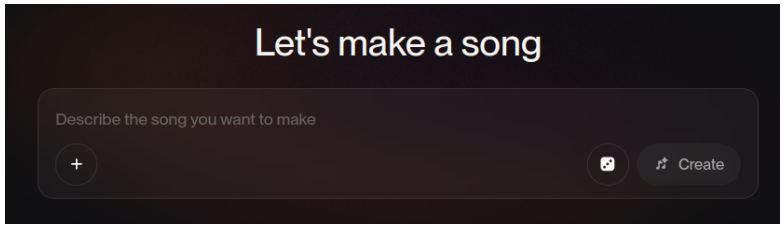
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Data and model: the example of Suno



"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

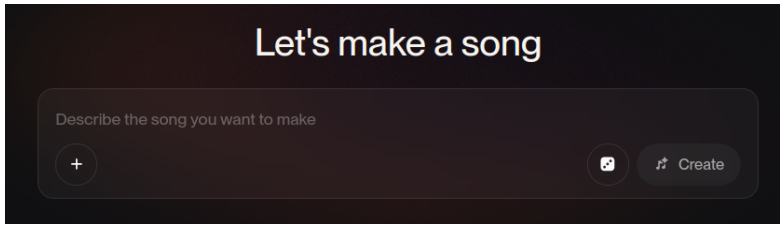
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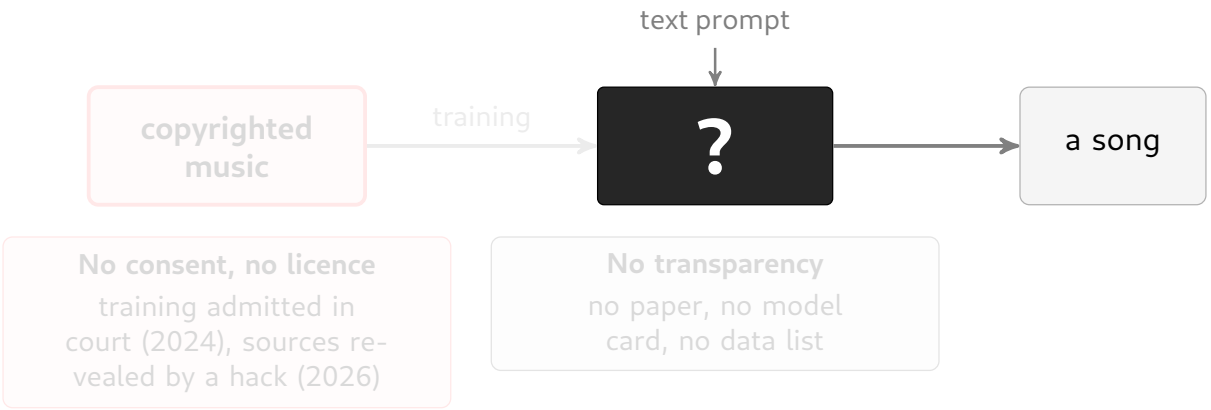
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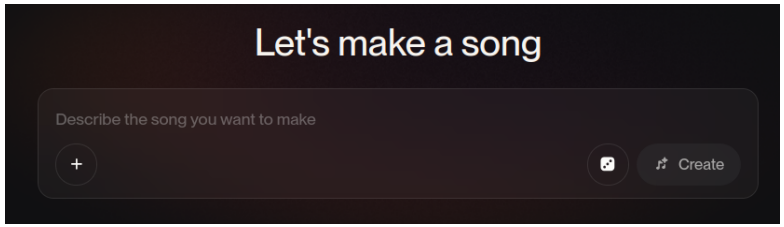
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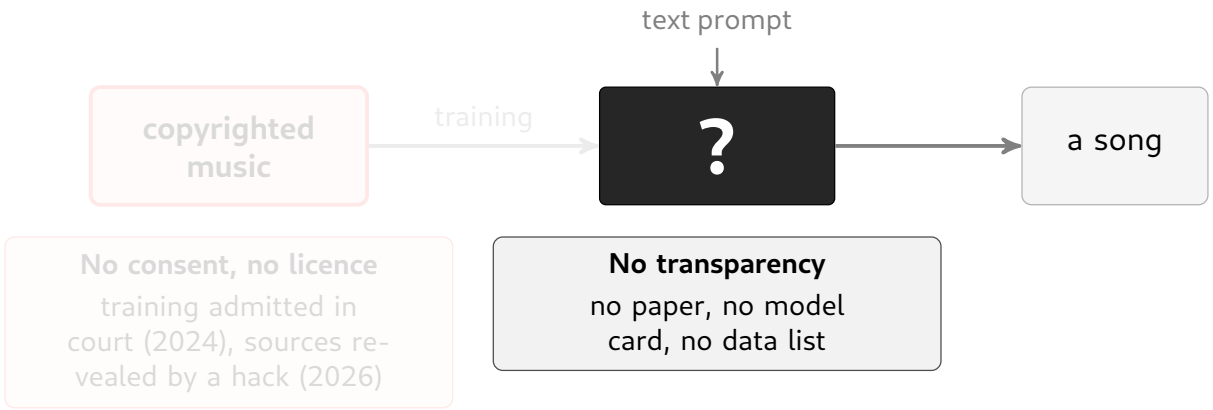
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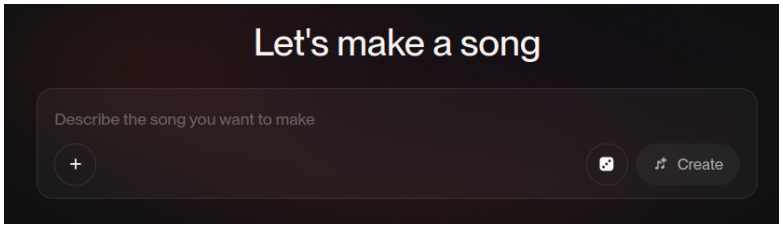
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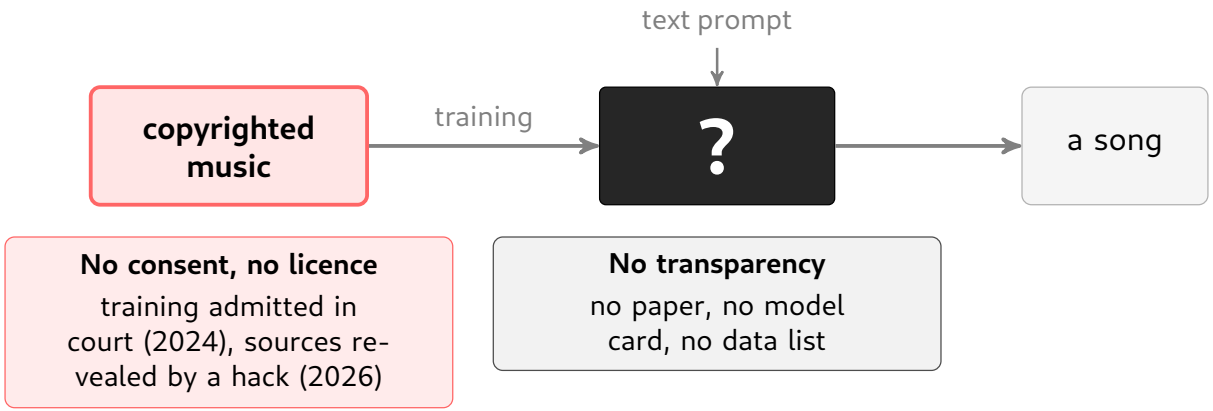
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Data and model: the example of Suno





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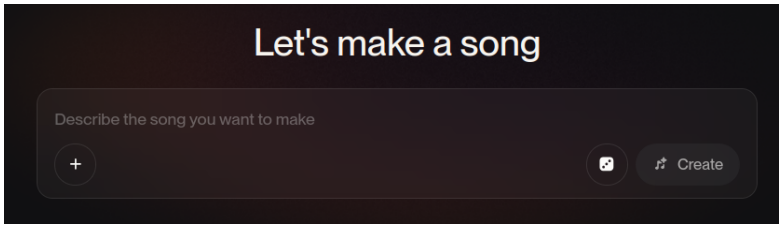
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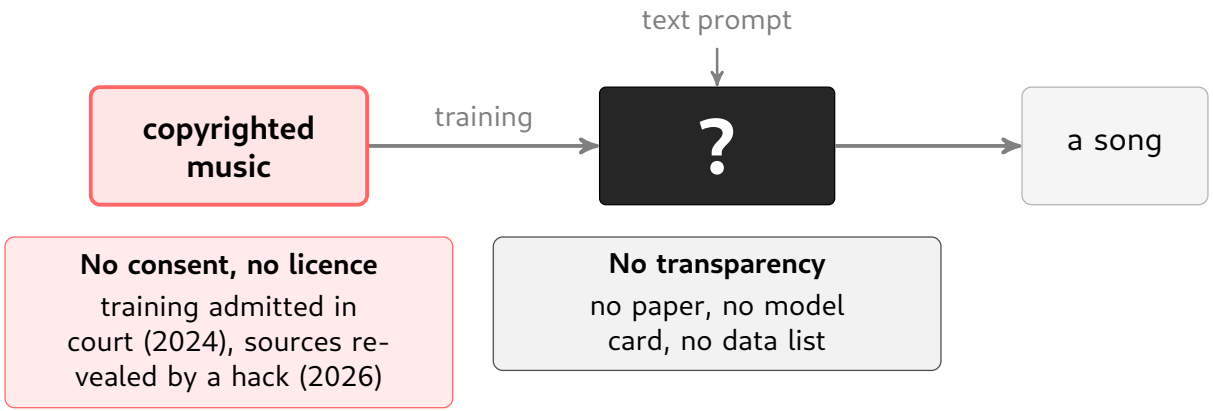
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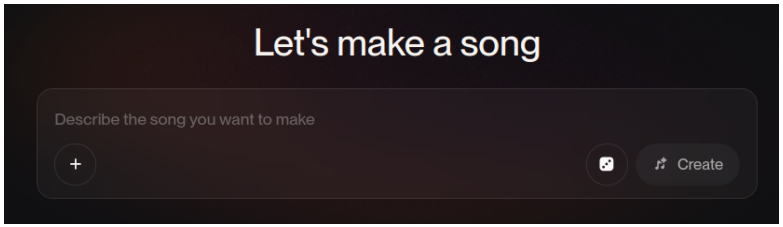
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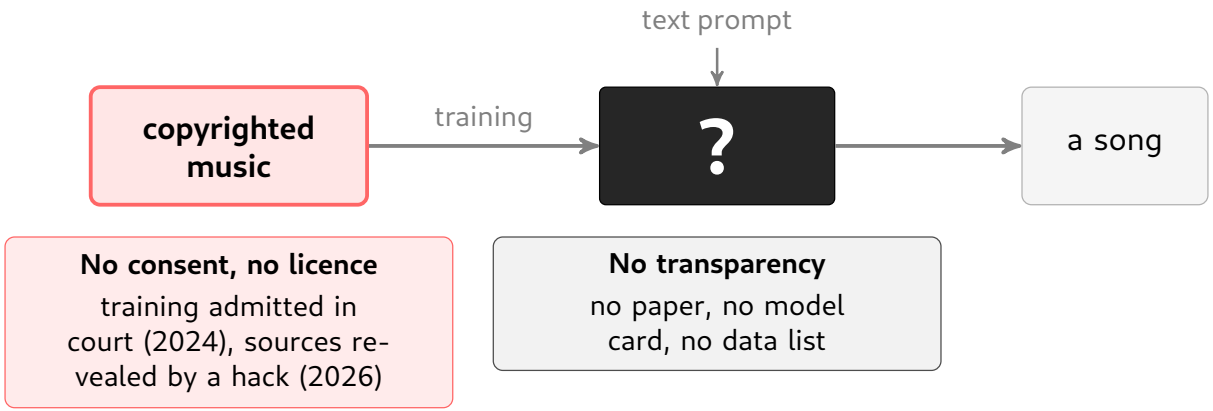
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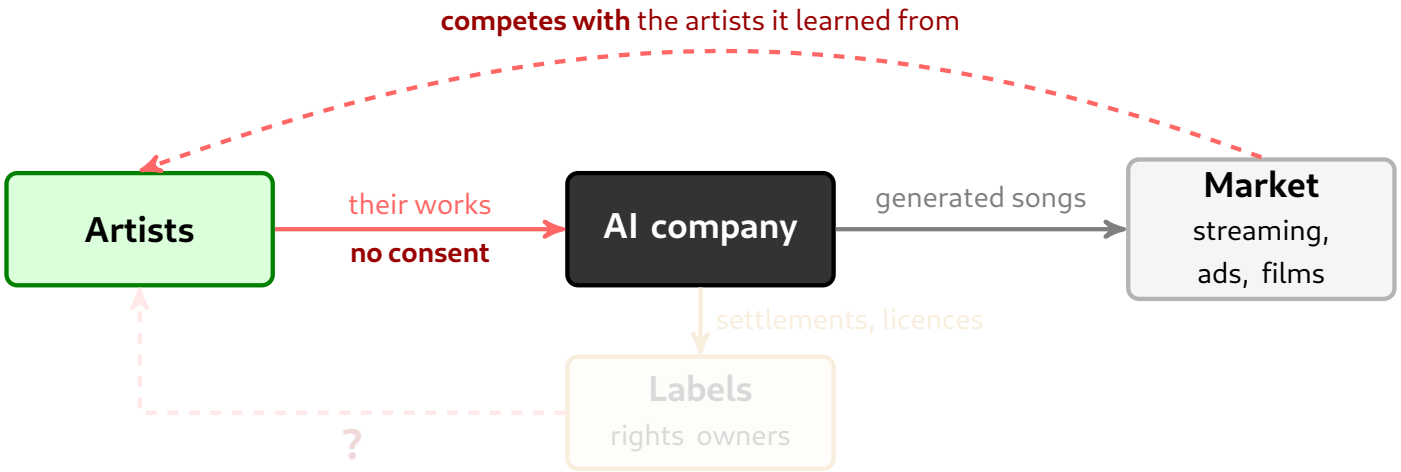
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Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

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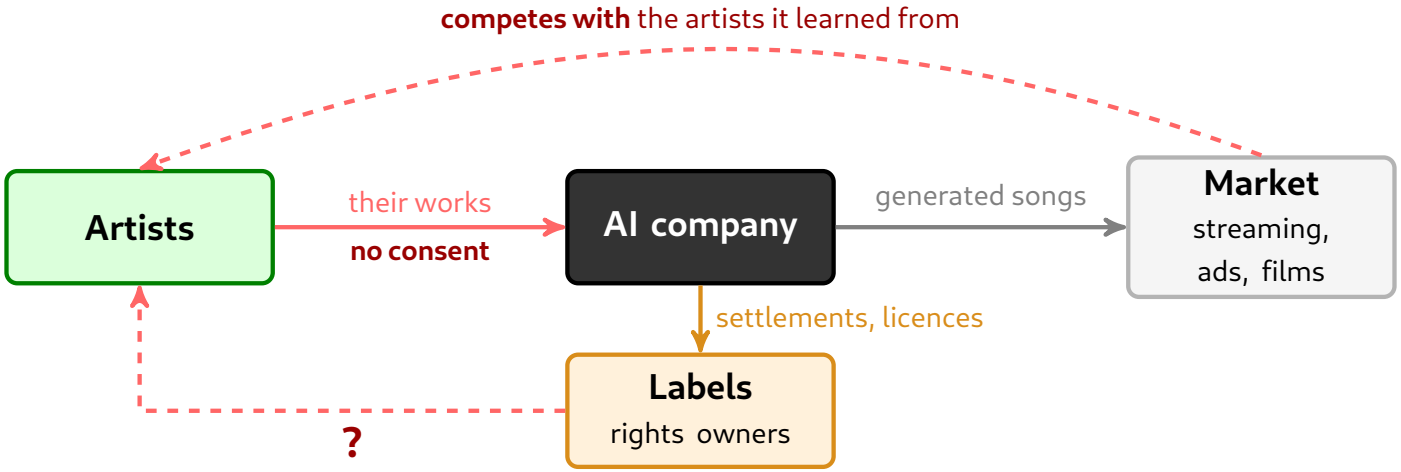
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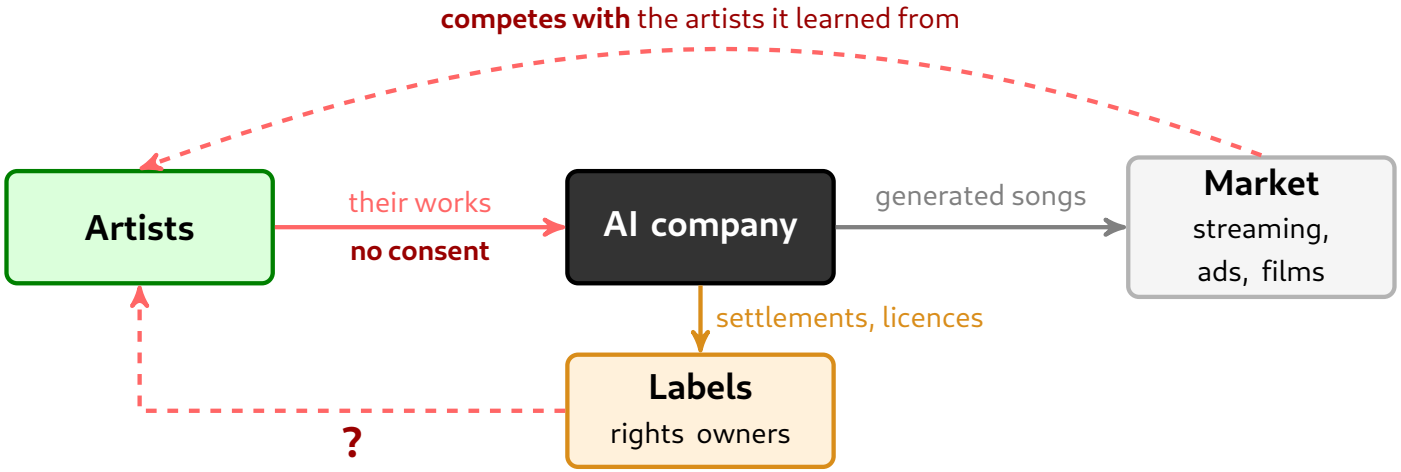
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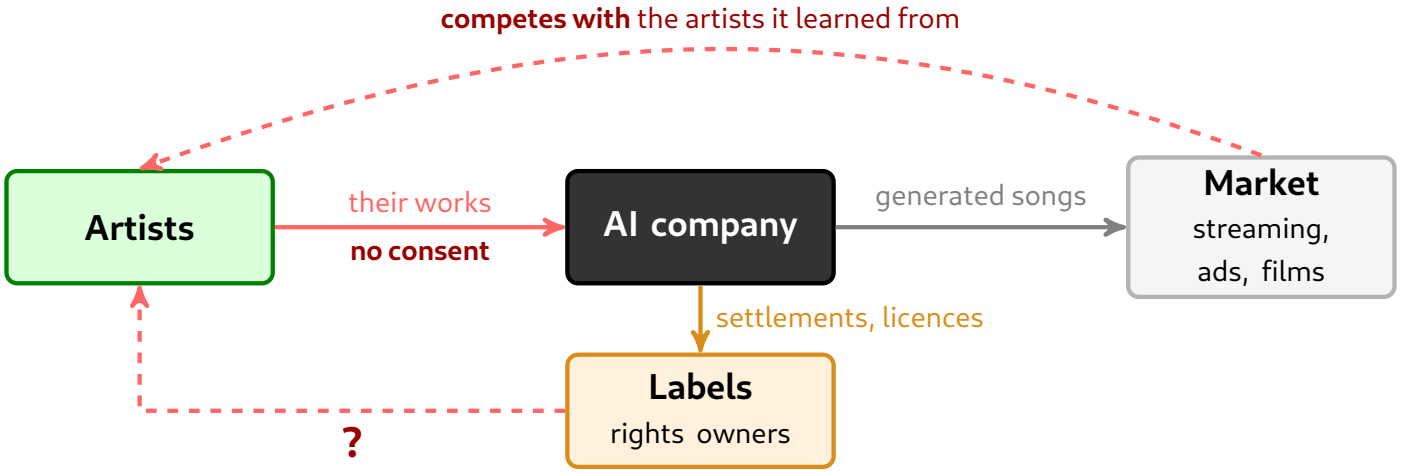
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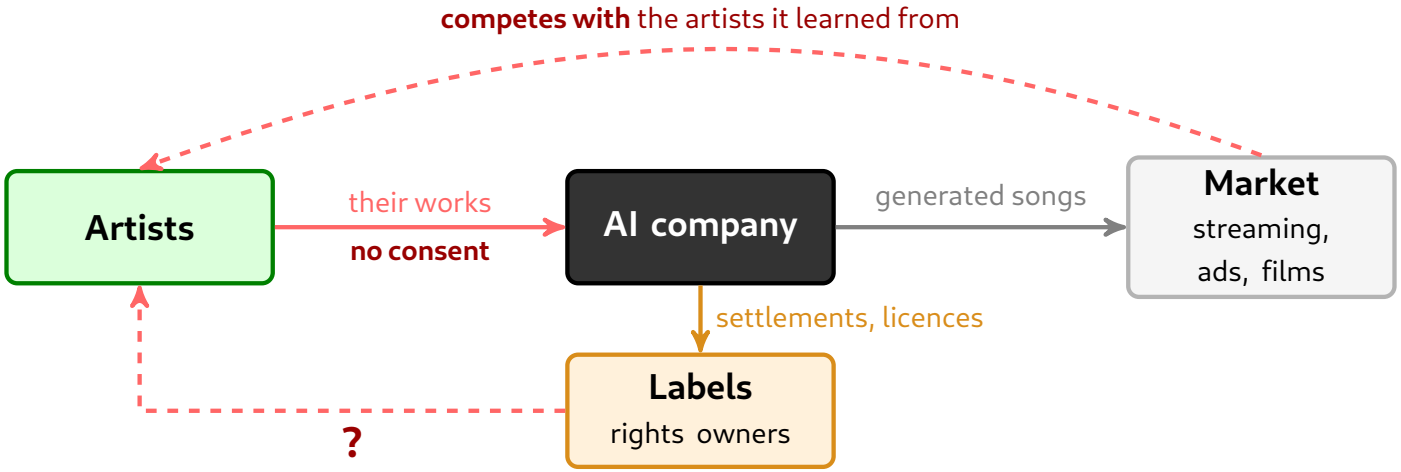
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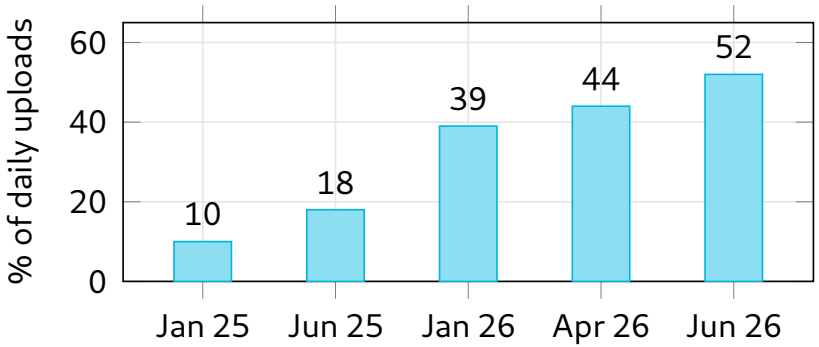
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The platforms are already flooded with AI music!

Flooded: fully AI generated uploads, Deezer



≈ **90,000 tracks a day** (Oct 2026)

company data, one platform: not generalizable

Barely listened: only **1 to 3%** of all streams

of which up to 85% bots (2025) | humans

bots stream the tracks to collect royalties: **fraud**

cause, correlation, or neither?

Deezer filters (own detector): AI tracks **excluded** from recommendations and playlists, bot streams **not paid**

Some AI bands make it

The Velvet Sundown (2025): ≈ **1M** monthly listeners on Spotify, then revealed as AI

97% fail a blind test (human or AI?)

80% want AI music labelled

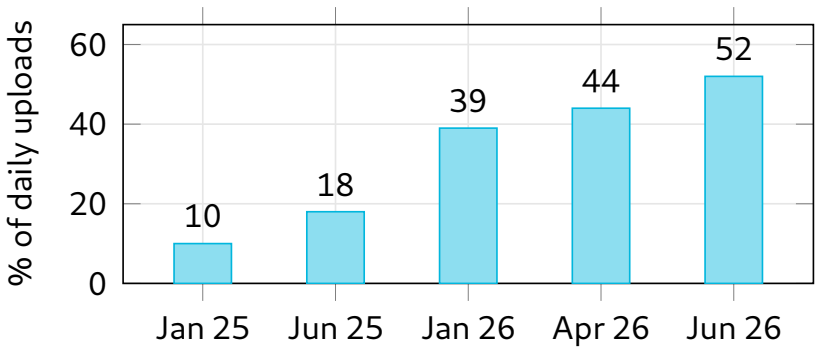
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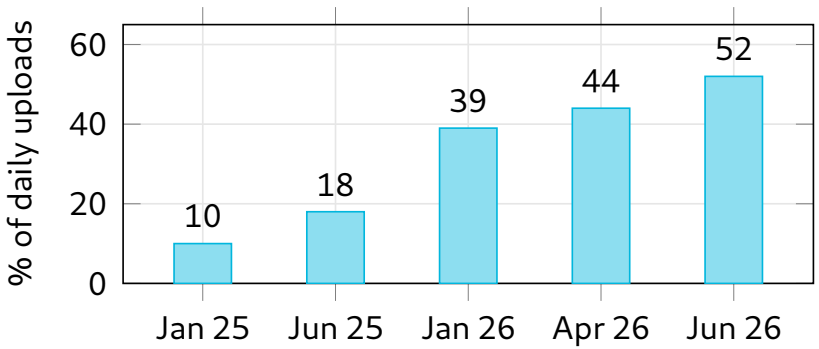
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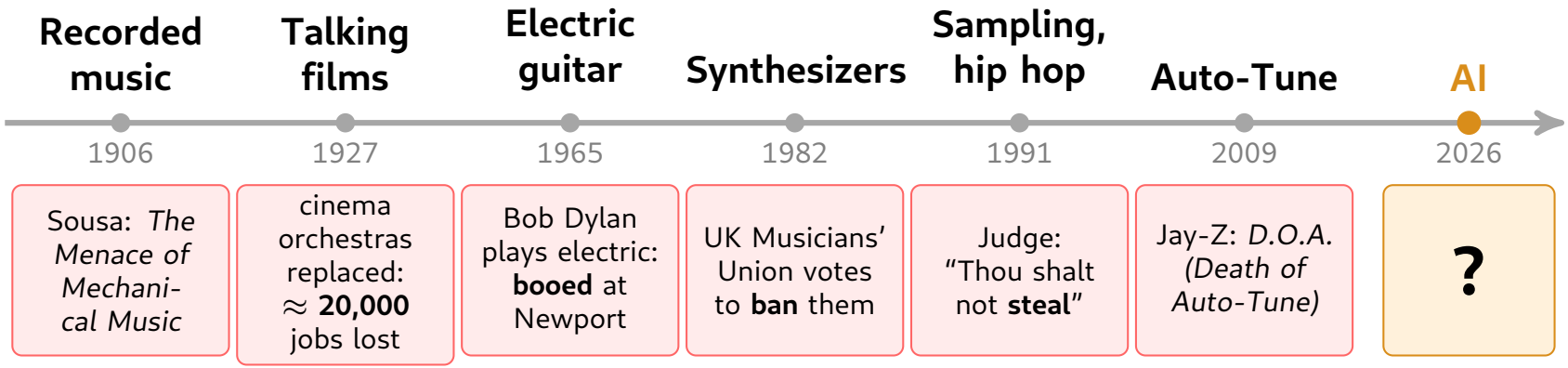
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Is AI the problem, or the next instrument?



first seen as a threat to music ⟶ all mainstream today
but "mainstream" does not mean "nobody lost": some fears were right

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After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

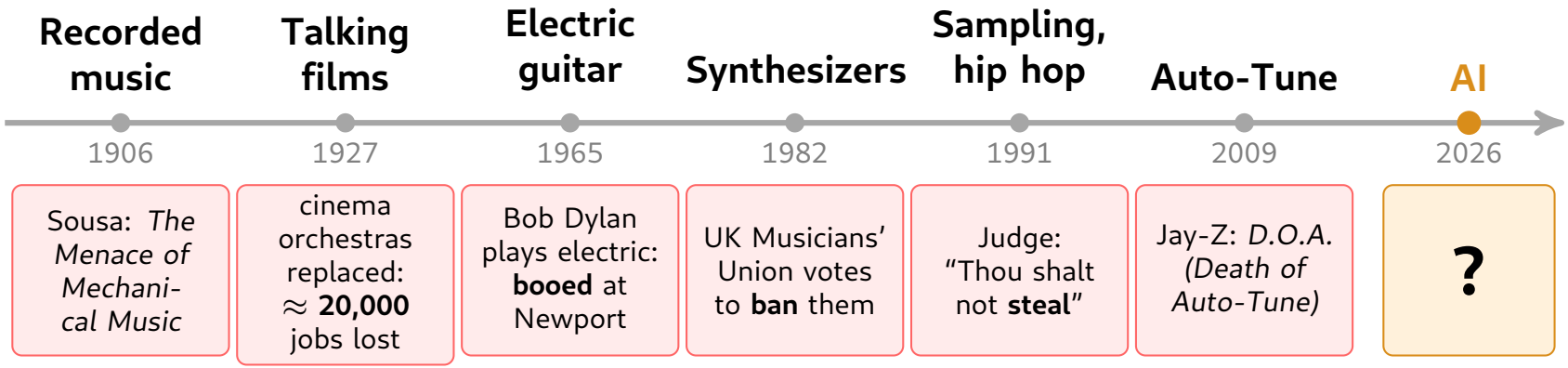
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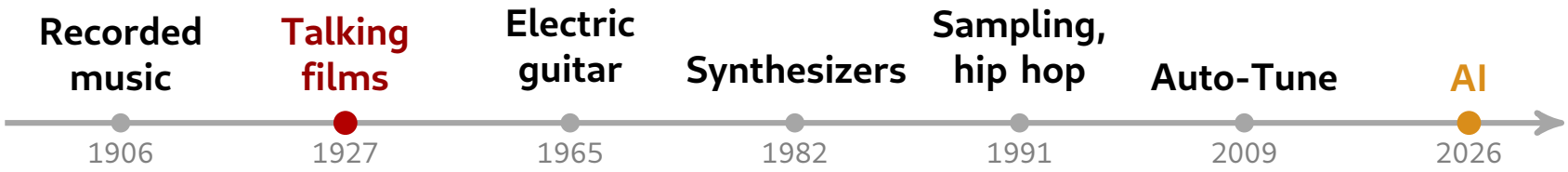
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1906	1927	1965	1982	1991	2009	2026
Sousa: *The Menace of Mechani-cal Music*	cinema orchestras replaced: ≈ **20,000** jobs lost	Bob Dylan plays electric: **booed** at Newport	UK Musicians' Union votes to **ban** them	Judge: "Thou shalt not **steal**"	Jay-Z: *D.O.A. (Death of Auto-Tune)*	**?**

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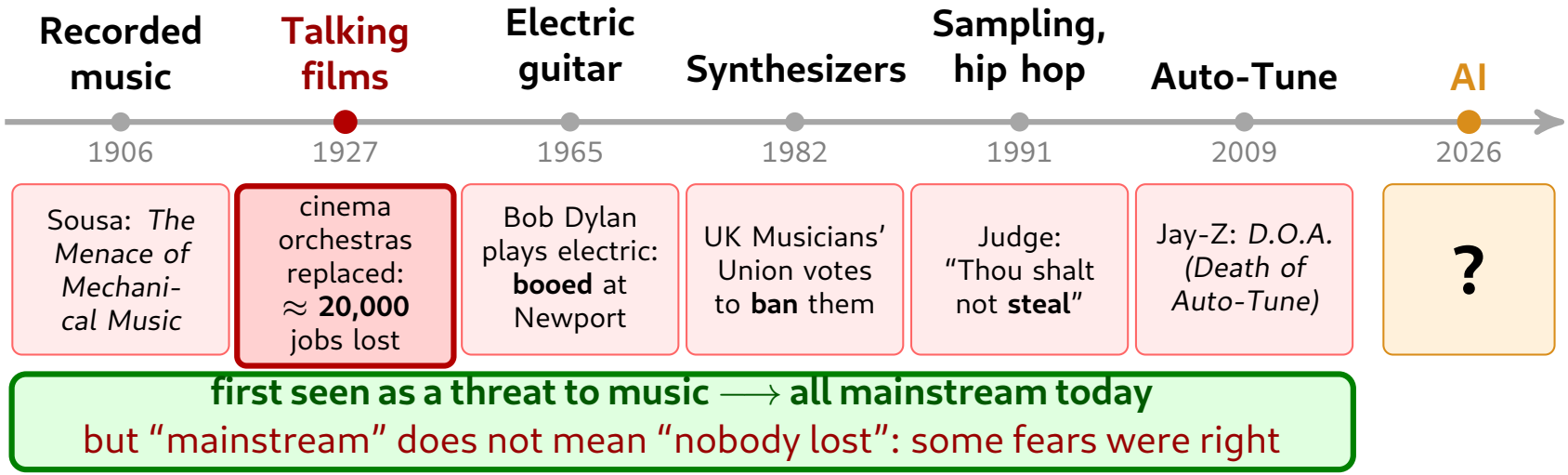
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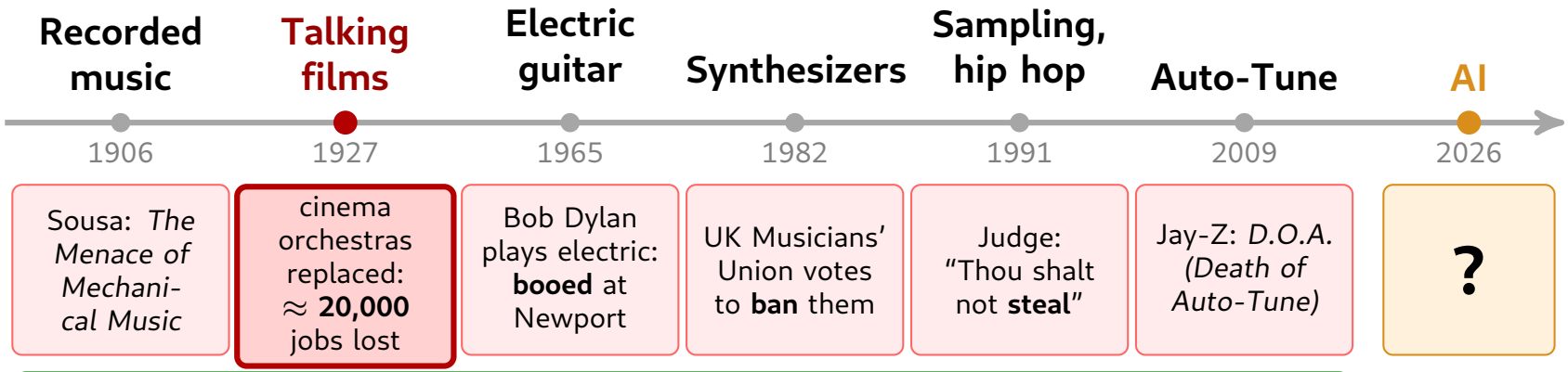
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Place yourself: Generative AI: Instrument, or parasite for the artists?

pick one side

INSTRUMENT *or* **PARASITE**

Strongest for "instrument"

- Access: **2M** paying users, and tools for musicians,
- It restores and **gives voices back**,
- Every music technology was first a **threat**; sampling got rules,
- Humans learn from music too; UK court: weights hold **patterns**, not works.

Strongest for "parasite"

- "Essentially all music" as data, **no consent**, in a black box,
- Munich: **memorisation**, GEMA wins against Suno,
- Money to labels; creators risk losing **24%** of their revenues by 2028 (projection),
- **52%** of uploads, up to **85%** of their streams bots.

Getty Images v. Stability AI, [2025] EWHC 2863 (Ch) GEMA v. Suno, LG München I, Jul 2026 CISAC and PMP Strategy, Dec 2024 other sources: previous slides

Outline

1 Agentic AI: Will AI destroy jobs?

2 Environment and sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate

Two sessions: prepare, then debate

1. **Work.** Will AI destroy jobs?
   yes ↔ no
2. **Environment and sovereignty.** Should France build its own frontier model?
   yes ↔ no
3. **Artistic expression.** Generative AI: Instrument, or parasite for the artists?
   instrument ↔ parasite

Session 1: become an expert

- You prepare **one** question, **on your own**,
- Read the slides, their notes, **and their sources**; look for new ones,
- Prepare arguments for **both** sides,
- On your question, you have **priority** at the start of the debate to introduce the first arguments.

Session 2: the moving debate

- **20 minutes** per question,
- The room is split in two: one side per answer,
- Experts open, then everyone joins,
- You **move** when an argument convinces you.

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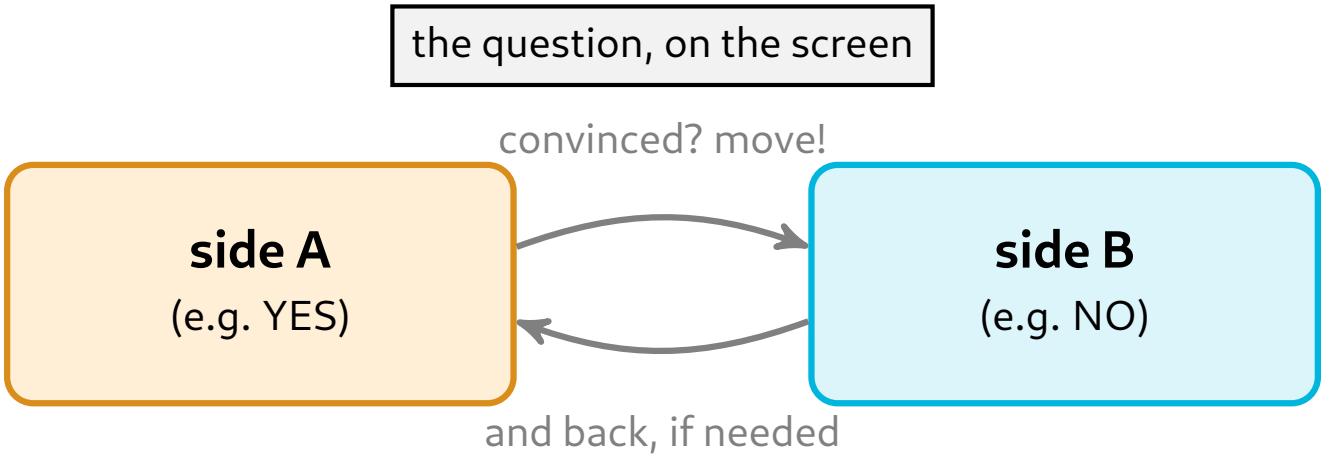
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The rules of the moving debate



Speaking

- **Everyone speaks at least once**, during the session,
- **One side after the other**: an argument for A, then one for B,
- **Share the speaking time**: be short, leave room for others.

Listening

- **Listen** to the end, without interrupting,
- **Kindness**: criticise the argument, never the person,
- Answer the argument that was given, not another one.

It is a *moving* debate

Your position is where you stand. If an argument convinces you, **change sides**, physically. Changing your mind is not losing: it is the point of the exercise.